

Makhura Kamogelo

Week 9 practicals

Tasks:

Part A: Multicast

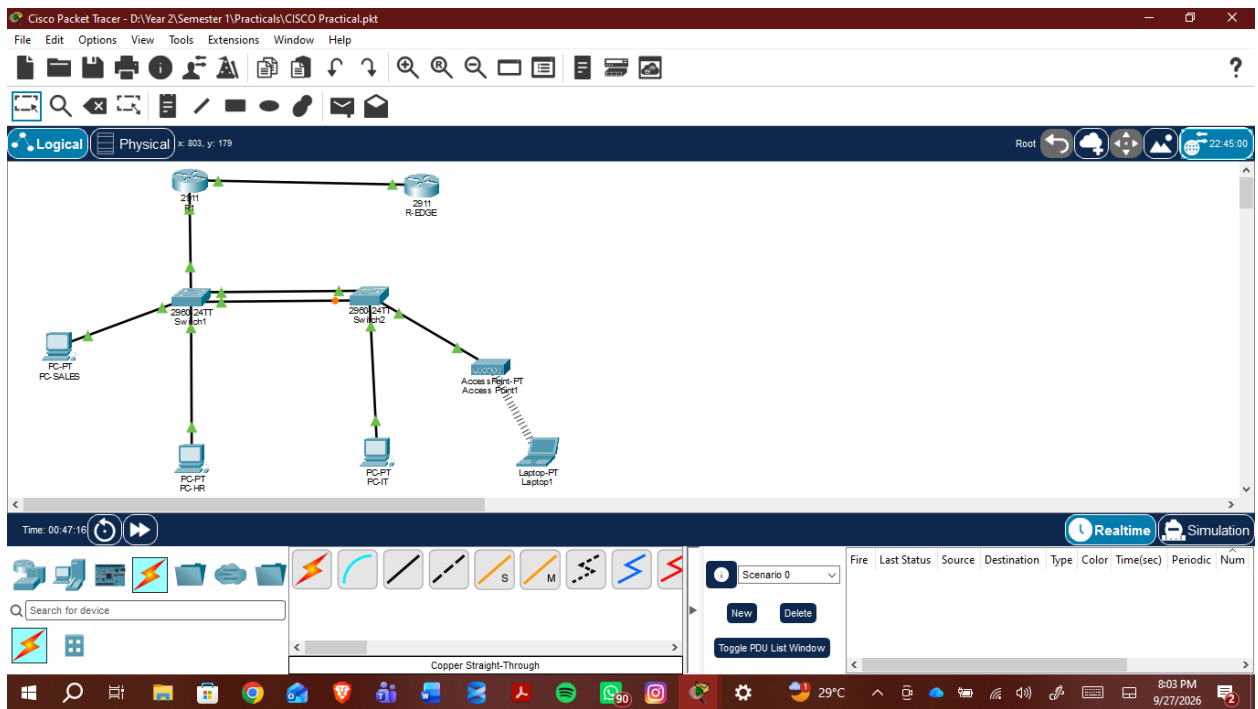
1. On relevant router interfaces: ip pim sparse-mode
2. Have a host join a multicast group (e.g. via a multicast video/test application), then verify: show ip igmp groups

Part B: QoS

1. Define a class-map and policy-map prioritising a chosen traffic type (e.g. voice/video) over general data.
2. Apply the policy to the relevant interface, then verify: show policy-map interface

Part C: Wireless

1. Configure a WPA2-secured SSID on the AP, integrated with an existing VLAN.
2. Connect a wireless client and confirm it authenticates and receives an IP address.
3. Deliberately misconfigure the passphrase on a second connection attempt and document the failure.



Cisco Packet Tracer - D:\Year 2\Semester 1\Practicals\CISCO Practical.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x 721, y: 328

Time: 00:47:21

Realtime Simulation

Scenario 0

Fire Last Status Source Destination Type Color Time(sec) Periodic Num

New Delete

Toggle PDU List Window

Copper Straight-Through

Laptop1

Physical Config Desktop Programming Attributes

Link Information Connect Profiles

Below is a list of available wireless networks. To search for more wireless networks, click the Refresh button. To view more information about a network, select the wireless network name. To connect to that network, click the Connect button below.

Wireless Network Name	CH	Signal
Default	1	53%

Site Information

Wireless Mode: Infrastructure

Network Type: Wireless B/G

Radio Band: Auto

Security: DSSS

MAC Address: 0021:433A:99ED

Refresh Connect

Adapter is Active

Wireless-N Notebook Adapter Wireless Network Monitor v1.0 Model No.: WPC300N

Top

Reflection Questions (answer these in your submission):

1. Why does QoS matter more on a network carrying voice/video traffic than on one carrying only file transfers?

Voice and video are real-time and delay-sensitive. Packet loss or jitter causes choppy audio/video. File transfers can retransmit and tolerate delay, so they don't need priority. QoS ensures bandwidth and low latency for voice/video.

"Packet Tracer version on 2911 router does not support ip multicast-routing command.